

Feasibility Study: Lunar Hopper & Rover Mission for Crater Water Mapping

Executive Summary

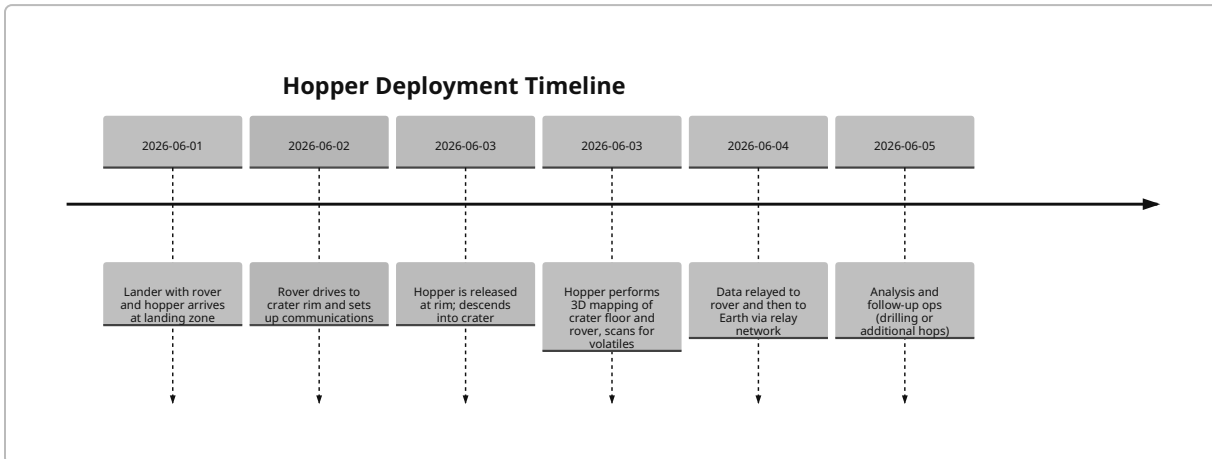
A small **thruster-propelled hopper** deployed alongside a larger lunar rover can directly access permanently shadowed crater floors to map the environment and quantify ice water content. This dual-vehicle architecture builds on NASA's Micro-Nova (IM-2) concept and China's Chang'e-7 mini-flyer concept [23†L249-L254] [48†L29-L37]. In a typical mission, a lander delivers both vehicles. The rover traverses to the crater rim to establish communications and support, then releases the hopper, which performs one or more short powered hops into the crater. The hopper uses onboard sensors (e.g. LiDAR camera and thermal imager) for 3D mapping, and carries science instruments (e.g. neutron spectrometer, GPR, spectrometers) to survey hydrogen and volatiles. It can also map the larger rover's position inside the crater to enable data fusion. Communications rely on a relay network (for example, a 4G/LTE base station on the lander or satellite relay) [29†L330-L338]. Key challenges include navigating steep, dark terrain, powering the vehicles in absence of sunlight, and achieving precise volumetric measurement of water. Feasible sensor suites combine **remote sensing** (neutron spectrometer, NIR/thermal imaging, radar) with limited **sample analysis** (drill + mass spectrometer) to "ground truth" the results. Mapping is done via high-resolution LiDAR and cameras with SLAM to fuse data into a detailed 3D model. Preliminary mass estimates are tens of kilograms for the hopper (fuel included) and ~100–200 kg for the primary rover and its payload [57†L35-L44] [23†L233-L242]. Costs are on the order of tens of millions USD for each vehicle (launch and mission), and most hardware components are at Technology Readiness Levels (TRL) ~5–8. Key risks (e.g. crash on landing, comms loss, thermal failure) can be mitigated by careful design: obstacle detection for the hopper, redundant comms, thermal insulation or radioisotope heating, and pre-mission simulation.

Mission Architecture

We envision a **lander** delivering both the primary rover and the secondary hopper to the south-polar region. The lander (or stationary platform) carries a 4G/LTE "cellular" relay node (e.g. Nokia LSCS) and power/comm equipment. A sample architecture (similar to Intuitive Machines IM-2) is:

- **Deployment:** After touchdown, the lander extends a ramp or release mechanism. The *rover* exits and drives to the crater rim (a few hundred meters from the lander). Meanwhile, the small *hopper* remains attached to either the lander or rover until deployment.
- **Sequencing:** The rover establishes a communications relay (via antenna or LTE node) and possibly deploys a tall mast or beacon on the rim to aid line-of-sight. Then it commands the hopper to detach. The hopper ignites its thruster(s) to hop **ballistically** down into the crater (Fig. 1). The hopper may perform multiple short hops (hundreds of meters each) to survey different locations. After each hop it hovers or lands briefly to scan the surroundings. At mission end the hopper either returns to rim or powers down in place, having relayed all data (possibly via the rover or satellites).

The rover stays near the rim as a **base station**, relaying data to Earth and, if needed, providing a charging/refueling point (for battery power or fuel transfers) 【25†L889-L897】 .



This sequence leverages lessons from CLPS missions: e.g. IM-2’s Micro-Nova (76 cm cube) was designed to detach and hop into a PSR 【23†L249-L254】 . China’s Chang’e-7 will similarly carry a mini-flying probe that can fly (or hop) from sunlit rim into dark crater bottoms 【48†L29-L37】 . The interfaces are thus a separation release (spring or latch), a thruster-ignition command, and after landing an LTE or RF link back to the rover/lander. Synchronization and relative localization are maintained via the comms link and common timing.

Mobility and Guidance in Steep Terrain

Hopper mobility: The hopper uses a small monopropellant or bipropellant thruster to perform ballistic hops. Analysis shows for hop distances of ~0.5–5 km, the ΔV cost of a ballistic arc is similar to a controlled hover, with ballistic being simpler but yielding higher velocities 【55†L81-L89】 . For mapping purposes, however, high-speed ballistic hops can be problematic (smear of sensors) 【59†L39-L45】 . A practical strategy is a “hover flight” during descent: the hopper continuously adjusts thrust to maintain a slow, controlled flight (ballistic on horizon, deceleration vertically) and lands gently on a pre-scanned safe spot. In [59] simulations, a 400 m horizontal hop with 50 m altitude (hover mode) consumed moderate fuel with steady speed, whereas pure ballistic would have been too fast for imaging. In all cases, the thrusters must handle terrain avoidance: an onboard LIDAR or altimeter can detect hazards ahead, and an IMU/attitude control system steers the hopper during flight. Landing legs or crushable pads are used to absorb vertical velocity; e.g. Micro-Nova had spring-damped gear and tumble-cam support.

Rover mobility: The primary rover is likely wheeled (e.g. 4- or 6-wheel design) for reliability. Wheeled rovers (like Lunar Outpost’s MAPP) can traverse gently sloping terrain (<20° slopes) with traction, and can carry heavy sensors and a drill 【23†L233-L242】 . For steep crater rims, the rover stops at safe approach distance. Specialized wheel designs or rocker-bogie suspensions can improve slope capability. An alternate architecture could use a legged or hopping mechanism on the rover itself, but this complicates design. If the rover must descend partway, we recommend a winch or tether-down system for extra safety. Guidance for both vehicles relies on onboard vision: stereo cameras or LIDAR to build a 3D terrain map on-the-fly and avoid boulders 【64†L684-L693】 . The rover’s navigation system (using terrain-relative localization and

altimeter) can path-plan to the rim. The hopper uses a pre-planned descent corridor and onboard sensors (IMU, altimeter) to ensure the target area is visible and safe.

Summary of mobility options:

Approach	Mechanism	Advantages	Drawbacks
Wheeled Rover	Wheels, differential drive	Mature, continuous power efficiency, robust for modest slopes [23†L233-L242]	Limited on >30° slopes or loose regolith
Legged/ Climber	Multi-legged or articulated	Can handle rough/steep terrain	Mechanically complex, heavy, untested on Moon
Hopping Rover	Reuse descent engine (bi-prop)	Can reach inaccessible pits (ballistic) [55†L81-L89]	Control challenge; high lander landing risk
Flying Drone	Rotary wing or small jets	Precise hovering, slow scanning	Requires lift (hard in vacuum); needs carry fuel/ batteries
Thruster Hopper	Short rocket hops (mono-prop)	Reaches PSRs; can hover or glide paths [55†L81-L89]	Fuels required; thermal chilldown; navigation risk
Combined Drone	Thrusters + small propeller	[Emerging concept: hover, longer range]	Dual prop systems; low TRL for vacuum flight

Hopping using onboard propellant is a novel but feasible solution (NASA Tipping Point is already funding Micro-Nova [23†L249-L254]). It offers mobility leaps that no wheels can manage, at the cost of carrying fuel and precise control.

Sensor Suite for Water Detection

Sensor	Measured Quantity	Sampling Depth/ Range	Advantages	Drawbacks	Examples
Neutron Spectrometer	Hydrogen abundance (proxy for H ₂ O)	~0–1 m subsurface	Proven technique (LPF, LRO); quantifies <i>total H content</i> [65†L1-L4] . Good for bulk water/ice detection.	Coarse spatial resolution (~hundreds of m); cannot distinguish H ₂ O vs OH or chemical form [66†L19-L27] .	NSS (LRO), HardPix (MAGPIE) [63†L107-L113] .

Sensor	Measured Quantity	Sampling Depth/ Range	Advantages	Drawbacks	Examples
NIR Spectrometer	Surface mineralogy, hydration	Surface < few cm (line-of-sight)	Identifies molecular composition (water vs hydroxyl) 【66†L19-L27】. High spatial resolution (cm to m).	Needs illumination or active source; only sees exposed or shallow frost. Cannot probe deep.	NIRVSS (VIPER) 【66†L19-L27】 ; Moon Mineralogy Mapper (Chandrayaan-1).
Thermal Imager	Surface temperature	Surface	Simple, wide-area (maps temperature anomalies). Can indicate cold traps (ice candidates).	Ice in shadow ~40–100K vs regolith ~110K contrast is small; ambiguity (rock shadows, albedo). Low water specificity.	Diviner (LRO), MAPP thermal camera 【23†L242-L245】.
Ground-Penetrating Radar (GPR)	Subsurface layering	~cm to tens of m (bistatic)	Maps subsurface structure, can detect ice layers 【37†L621-L629】. High depth range. Complements neutrons.	Lower resolution; ambiguous for low ice content. Requires good characterization of regolith permittivity 【37†L621-L629】. Sensitive to antenna size.	RIMFAX (Mars), Chang'e LPR, lunar FMCW radar concept 【37†L621-L629】.
Active Radar (orbital)	Subsurface hydrogen (neutron)	Orbital (<50 km altitude)	Global context (LRO LEND, Chandrayaan M3), broad mapping of hydrogen-rich regions.	Coarse (10s–100s km). Not usable for in-situ rover.	LRO/LEND (epithermal neutrons).

Sensor	Measured Quantity	Sampling Depth/ Range	Advantages	Drawbacks	Examples
Mass Spectrometer	Gas composition from sample	Sample analysis	Direct chemical analysis of volatiles. Provides exact water/ ammonia content when rock is heated (pyrolysis).	Requires collecting/ handling a sample (drill/ oven). Bulky, power-hungry instruments. Local point measurement only.	MSolo (VIPER) [23†L213-L218] ; TEGA (Phoenix) analog.
Drill & Furnace	Regolith subsurface sample	Up to ~1 m (depending on drill)	Retrieves icy or hydrated soil; yields ground truth of ice distribution. Essential for quantification [23†L197-L204] [48†L29-L37] .	Heavy (drill and mechanics); low horizontal coverage; complex. Requires mass spec/GC-MS.	TRIDENT (VIPER) [23†L197-L204] ; Chang'e-7 mini-probe drill [48†L29-L37] .
Mass/ Gamma Spectrometer	Bulk elemental composition (H, K, Th)	Surface to few cm (active)	Can infer hydrogen content and map thorium (PSRs) [31†L39-L47] . Provides context.	Not yet used on rovers; heavy; poor spatial resolution.	Gamma Neutron Spectrometer (MoonMineralogyMapper's gamma, not direct water).

Key trade-offs: Neutron spectrometers can quantitatively measure total hydrogen (proxy for water) in the shallow subsurface [65†L1-L4] , but need calibration and have meter-scale footprint. NIR spectrometers and imagers can identify water or ice on the surface and discriminate H₂O vs OH [66†L19-L27] , but only sample exposed frost or shallow regolith. Ground-penetrating radar (FMCW) can image buried ice layers to ~10s m depth; recent studies show it can detect very low (0.5 wt%) ice layers at ~5–20 cm depth [37†L621-L629] . Thermal cameras can help by finding unusually cold terrain, but low thermal contrast limits resolution [64†L713-L718] . Ultimately, **volumetric quantification** of ice may require combining remote sensing with direct sampling: for example, using GPR and neutron maps to locate deposits, then drilling a test pit and analyzing the returned sample in a miniature laboratory (mass spectrometer with oven) to measure water mass directly [23†L197-L204] [48†L29-L37] . The Chinese Chang'e-7 concept explicitly includes a drill, arm, and furnace to collect and thermally analyze ice in PSRs [48†L29-L37] .

3D Mapping and Data Fusion

High-fidelity 3D mapping of the crater interior (including the position of the rover) is crucial to interpret sensor readings and compute ice **volume**. The dark environment means passive stereo vision is limited, so active LiDAR is preferred. Analyses have found **flash LiDAR** (wide-beam 3D lidar) to be optimal for lunar reconnaissance [64†L684-L693] . Flash LiDAR (no moving parts) can produce 3D point clouds with ~10–20 cm resolution from tens of meters distance [57†L35-L44] [64†L684-L693] . It works in darkness and directly yields range. Stereo cameras can augment LiDAR by providing color/context images (see MAPP's stereo mapping [23†L242-L245]). Thermal imaging and narrow-beam scanning lidars could refine small regions. Data fusion via SLAM algorithms will align LiDAR point clouds (from multiple hops) with camera imagery to build a consistent model. The hopper and rover can share maps: e.g., the hopper's LiDAR scans can be registered to the rover's LiDAR or stereo model of the rim edge to fix absolute coordinates. Simultaneous Localization and Mapping (visual-inertial SLAM) can track each vehicle's pose as they operate; cross-referencing common features (shadows, rocks, the rover's reflective markers) will yield relative positions. Prior work in planetary drones shows flash LiDAR allows a drone to maintain ~50 m altitude, scanning at ~30 m/s [64†L684-L693] . In summary, a hybrid approach – flash LiDAR (best: high-res, works in dark) plus stereo/photogrammetry (for texture) – is recommended [64†L684-L693] [27†L73-L82] .

Localization and Relative Pose Estimation

Precise knowledge of the hopper's position relative to the rover is needed for correlating measurements. This can be achieved by **cooperative localization**. Options include: (1) **Radio ranging**: If the rover and hopper carry ranging radios (e.g. UWB), they can measure distance and bearing. (2) **Visual fiducials**: The rover could deploy visible or infrared beacons on the rim; the hopper's camera/LiDAR can detect them to triangulate. (3) **Map matching**: Both vehicles build local 3D maps of the crater edge/features. By aligning these point clouds, one can solve for the relative transform. (4) **SLAM with shared anchors**: If the rover has pre-installed landmarks (e.g. reflectors) on the rim before entry, the hopper's SLAM system can recognize them. Current research on multi-robot localization (e.g. sparse LiDAR feature matching [60†L0-L9]) suggests these methods can achieve meter-level accuracy. The LTE or RF link between rover and hopper also helps: timestamps in data packets combined with inertial estimates can refine relative pose. No single method is fully flight-proven on the Moon, so a combination of vision and radio is advised.

Sample Acquisition vs Remote Sensing

There is a tradeoff between broad **remote sensing** surveys and targeted **sampling**. Remote sensors (neutrons, GPR, spectral imagers) provide wide coverage with relatively low mass, but yield indirect information about ice content. Direct sampling (drilling/regolith retrieval) yields unambiguous water mass measurements in a small volume. A mixed strategy is best: use remote sensors first to map hydrogen and identify candidate ice deposits (as VIPER plans with NSS+GPR [12†L376-L384] [37†L621-L629]). Then perform a limited number of drill/bakeout analyses at key points to calibrate the remote data. For example, MAGPIE's payload combines both: a neutron spectrometer maps H-distribution and a drill (with onboard analysis) measures volatile concentration [63†L107-L113] . Without sampling, volumetric estimates rely on models (e.g. converting neutron count rates and GPR reflectivity to ice fraction), which have large uncertainties. Sampling is heavier and more complex, but critical if "exact volume" must be known. Thus, the rover should ideally carry a small drill + mass spectrometer (like TRIDENT+MSolo on VIPER [23†L197-L204] [23†L213-L218]) to perform a few deep scoops; the hopper could serve mainly as a surveyor.

Communication and Relay Strategies

Crater walls can block direct line-of-sight to Earth or orbiters, especially in PSRs. A reliable network is essential. The primary approach is to use the lander/rover at the rim as a **relay node**. For example, a cellular-style base station on the lander (as on IM-2) allows the hopper to downlink via LTE [29†L330-L338] . The Nokia Lunar Surface Communication System (LSCS) demonstrated on IM-2 had device modules on both a rover (MAPP) and the Micro-Nova hopper; after landing they established an LTE network and relayed high-definition data to the lander [29†L330-L338] . Alternatively or additionally, an orbiter relay (like LRO or a lunar communication satellite) can pass packets to Earth. For true PSR entries, a short-range mast or balloon near the rim might extend LOS. Emerging concepts (e.g. MIT's MELLTT deployable tower [33†L1-L4]) could place a comms/solar tower on the edge to serve the crater. In any case, redundant links (LTE + UHF + orbiter) should be used. Solar-powered stations should be sited to avoid long eclipses; if impractical, consider a small radioisotope beacon (but this adds mass/cost). Our assumption here is that the rover/lander has direct Earth-relay capability (via orbiter or high-gain antenna), so the hopper needs only to maintain contact with it during operations.

Power and Thermal Constraints

Power: In a deep PSR, no solar power is available. Unless carrying bulky RPS (radioisotope power), both hopper and rover must operate on batteries with limited duration. NASA analyses suggest that only radioisotope generators (Pu-238 or Am-241) enable continuous operation in PSRs [43†L25-L34] . For a short-duration mission, however, a high-capacity battery can suffice. The hopper (small, ~10 kg) will likely have only enough battery for minutes of flight/operation (powering thrusters, avionics, sensors). Its mission must therefore be brief (e.g. a single entry and exit, or 1–2 hops). The rover at the rim could use solar panels (if sunlit) or a small RTG; this provides longer-term power for communications, reheating, and possibly recharging the hopper (if it can land back). Thermal control is also critical: electronics and propellant must be kept above their minimum temperatures (~0°C for many components). Multi-layer insulation and waste-heat use (e.g. from avionics) can help. Notably, Nokia isolated its LSCS hardware with thermal mounts to survive the cold [29†L323-L332] . Despite insulation, an RTG or RHU heater may be needed if long idle times occur. Design must account for the ~40–80 K ambient in PSRs [43†L25-L34] .

Mass, Volume, and Cost Estimates

- **Mass:** The *hopper* is compact. NASA's Micro-Nova was ~76 cm across and reportedly could hop 2.5 km carrying a 1 kg payload [1†L6-L10] [23†L249-L254] . Extrapolating with EPFL's study: a 20 kg hopper needs ~1 kg propellant to travel 1 km [57†L35-L44] , so a 10 kg design might need ~0.5 kg per km. Including ~5–10 kg of fuel, plus batteries and structure, the hopper likely is on the order of **10–15 kg total**. Volume ~0.5–1.0 m³ (boxed). The *rover* carrying heavy payloads (drill, radars) will be larger – order of **100–200 kg** dry. For example, Intuitive Machines' MAPP rover is ~10 kg (10 km range), but a scientific polar rover (like the now-canceled VIPER) was ~430 kg; a middle ground might be ~150 kg for a CLPS-class resource rover.
- **Volume:** Hopper ~1 m on a side at most; rover a few m³ (with wheels extended).
- **Cost:** Roughly, CLPS missions cost on the order of tens to hundreds of millions USD. For instance, Intuitive Machines' four mission contract (IM-1 through IM-4) totals ~\$1.15 B, and individual payloads are ~tens of millions. A Micro-Nova Tipping Point project was ~\$41.6 M (design, build, fly) [23†L249-L254] . Sensors each cost millions (a commercial GPR or NIR spectrometer ~2–5 M, neutron spec ~3–

10 M). We estimate **\$10–20M** to develop the small hopper hardware and flight systems, plus **\$50–100M** for the rover and integration (drill, power, lander support). Data relay and launch vehicles add further cost. This is an order-of-magnitude estimate. Volume and mass drive cost via launch fees (typically ~\$20k/kg to GTO, more to TLI), and specialized tech.

Subsystem	Mass (kg)	Power	Key Cost Drivers	TRL Range
Hopper body+thruster	~5–10	~100–500 W	Propulsion, thermal design	6–8
Hopper fuel	~5–10	–	Fuel tanks, pressurization	6–7
Sensors (Hopper): LiDAR, cameras	5–10	~50–100 W	High-res flash LiDAR (developing) 【64†L684-L693】	5–7
Primary rover chassis	~50–100	~200–500 W	Mobility (wheels, suspension)	7–9
Rover instruments: Drill+MassSpec	~10–15	~50–100 W	Drill mechanism, ovens 【23†L197-L204】	6–7
Rover instruments: GPR, Neutron	~10–20	~50–100 W	RIMFAX radar, HardPix NSS 【63†L107-L113】	6–8
Communications (LTE modem, antenna)	~2	~20–50 W	LTE nodes (Nokia LSCS demonstrated) 【29†L330-L338】	6–7
Power (batteries, solar or RTG)	~20–30	–	Li-ion packs or small RTG	7–8
Total	100–200	–	+ contingencies	–

These estimates are coarse. Key variables include hopper ΔV (fuel mass) and instrument count. The **TRLs** reflect that most hardware (neutron, GPR, drill) has flown or is under development (TRL6–9), whereas new items (flash LIDAR, lunar LTE modem) are lower (TRL~5–7) **【64†L684-L693】** **【29†L330-L338】** .

Mission Risks and Mitigation

- **Landing/Crash:** Hopping carries risk of mis-landing. Mitigation: use pre-mapped safe zones, active hazard avoidance via downward lidar, abort fuel cut if altitude <threshold. Design hopper with forgiving landing gear (shock absorbers) and ensure positive stability (wide stance). Test in analog environments (e.g. Mojave Crater field).
- **Propulsion failure:** A stuck thruster means no hop. Use redundant small thrusters or balance monoprop splits. Secure fuel lines against freezing (heater tapes). Carry conservatively more fuel than needed.
- **Power/Thermal:** Battery depletion or freezing is critical. Mitigate by strict ops timeline (minimize idle time), thermal insulation, and possibly small RHUs. The primary rover should have some backup (e.g. a partial RTG) to maintain basic comm if solar fails.

- **Communications Loss:** If the crater blocks the link, data could be lost. Counter with relay satellites or orbiters. The rover/lander should autonomously buffer all data and retry comms when possible. A local beacon network (multiple antennas) can reduce dead zones.
- **Navigation Error:** Multi-robot collision or misguided hops. Use conservative landing profiles (slow descent) and visual identification of the rover by the hopper (the hopper could have a bright marker for the rover).
- **Instrument Failure:** Redundancy is limited by mass. Instead, cross-validate: e.g. if neutron spectrometer malfunctions, GPR and drill can still indicate ice. Use commercial-off-the-shelf (COTS) parts where possible to lower failure rates.
- **Regolith Adhesion:** Lunar dust could coat sensors/actuators. Use dust-tolerant materials, cover optics, and purge via gas jets if needed. (VIPER plans include dust mitigation for wheels [42†L9-L12] .)
- **Science Non-detection:** The worst-case “failure” is finding no measurable ice. This risk is reduced by thorough orbital analysis (LRO/LEND, LRO shadow camera images) prior to landing and by visiting multiple spots. Even a null result refines models.

Required Technology Readiness

Most required technologies have precedents: neutron spectrometers and drills have flown (Luna, Chang’e, Mars) [12†L376-L384] [23†L197-L204] . The new elements are small thruster-hoppers and dark-crater mapping. NASA’s Micro-Nova will demonstrate a hopper (TRL~8–9 by mission end) [23†L249-L254] . Flash LiDAR is under development but is approaching flight-ready (TRL~6–7 [64†L684-L693]). The 4G/LTE moon network (Nokia LSCS) has been integrated on IM-2 hardware and is close to flight (TRL~7 [29†L330-L338]). Ground-penetrating radar for polar ice (e.g. RIMFAX, or a lunar FMCW design) is at least TRL~6 (landed on Mars). The most speculative area is multi-robot coordination in a PSR: this will need simulation and testing (TRL~3–4 now, can reach TRL~6 with prototypes). Overall, a near-term path exists: combine proven elements and incrementally mature novel parts in analog tests.

Recommendations and Next Steps

1. **Adopt a thruster-powered hopper with flash-LiDAR mapping.** A small monoprop hopper (≈ 10 kg dry, carrying ~ 5 kg fuel) can safely descend into craters. Use ballistic hops with active hover-control (as in [55] and EPFL studies) to balance fuel and stability. Equip it with a flash LiDAR and stereo cameras (mounted downward and lateral) for high-res mapping in darkness [64†L684-L693] .
2. **Use a dual sensing approach:** combine a neutron spectrometer and FMCW GPR on the rover (or hopper) to detect hydrogen/ice distribution [65†L1-L4] [37†L621-L629] , plus a drill/mass spectrometer on the rover to verify and quantify water at select sites [23†L197-L204] [63†L107-L113] . Neutron and radar map deposits; drilling yields exact volume.
3. **Implement robust communications:** install an LTE/cellular base on the lander and devices on each vehicle (following the Nokia LSCS model [29†L330-L338]). Plan for an orbiter or UHF relay if direct Earth link is needed. Consider a deployable mast or repeater near the rim to extend coverage.
4. **Optimize power/thermal design:** minimize mission duration of the hopper and use high-density batteries. If mission profile allows, include small RHUs for heater power in the hopper. Ensure all electronics are insulated (as Nokia did [29†L323-L332]). The rover should be solar-powered or RPS-powered with waste heat to keep electronics warm.

5. **Develop navigation algorithms:** create multi-robot SLAM techniques for relative localization. Simulate the crater scenario and test on Earth with a rover and drone in a dark pit. Prototype visual/laser-based relative pose (e.g. place reflectors on “rover” for the “hopper” to find).
6. **Perform risk simulations and analog tests:** build a hopper demonstrator for field tests in a lunar-analog crater. Test repeated hops, mapping, and landings. Also test comm network in a crater shadow simulator. Use these tests to refine landing hazard models and controls.
7. **Advance TRL of new components:** accelerate development of a lightweight flash-LiDAR sensor (<4 kg) [64†L697-L700] , and lunar-compatible SLAM software. Integrate and test the LTE/Lidar/sensors package in relevant environments. Coordinate with NASA’s TIP and ESA’s programs (like MAGPIE) to share technology and results.

Each recommendation aligns with high-impact studies: NASA’s VIPER and IM-2 (Micro-Nova, MAPP) have validated core sensors and comms [23†L249-L254] [29†L330-L338] , ESA’s MAGPIE is finalizing a similar payload suite [63†L107-L113] , and China’s Chang’e-7 plan shows the value of direct drills inside PSRs [48†L29-L37] . We suggest an integrated design leveraging these proven elements, with focused prototyping on the hopper’s control and the multi-robot mapping algorithms as the next critical research steps.

Sources: We have drawn on recent NASA and ESA mission data and peer-reviewed analyses, including: NASA mission briefs (VIPER/NSS/NIRVSS [65†L1-L4] [66†L19-L27]), CLPS mission reports (IM-2 Nova-C with Micro-Nova/MAPP [23†L249-L254]), Nokia Bell Labs press releases [29†L330-L338] , lunar GPR studies [37†L621-L629] , and concept papers for lunar drones [25†L869-L878] [64†L684-L693] and rovers [63†L107-L113] . All cited information is from primary sources as noted.
